

Reproducing the DESI Strong Lens Foundry II Spectroscopic Catalog from Public DESI DR1

Greg Benson

Xiaosheng Huang

June 2026

Internal Reproduction Report

Abstract

We reproduce the spectroscopic backbone of the DESI Strong Lens Foundry II catalog (Huang et al., 2025b): the lens redshifts (z_d), source redshifts (z_s), and lens velocity dispersions (σ_v) for all 73 strong-lens candidate systems tabulated in their Table 2. Foundry II measured these quantities from the DESI Early Data Release (EDR, equivalently the `Fuji` production); we instead re-derive them from the *public* DESI Data Release 1 (DR1, the `Iron` production) products already on local disk — the 28.4M-fiber `zall-pix-iron` redshift catalog and the on-disk FastSpecFit DR1 v3.0 velocity-dispersion shards. Our parsed Table 2 reproduces the paper’s stated sample exactly (20 confirmed, 1 known, 4 nonlenses, 48 pending; 72 lens redshifts, 22 source redshifts, 71 σ_v values). Cross-matching to DR1, **all 73 systems** have a DESI fiber within $1.5''$; we recover **70 of 72** published lens redshifts and **16 of 22** source redshifts to $|\Delta z| < 0.005$ (median $|\Delta z| \approx 3 \times 10^{-5}$), and **65 of 71** velocity dispersions ($r = 0.80$, median ratio 0.96, median $|\Delta \sigma_v| = 22.5 \text{ km s}^{-1}$ for well-measured systems). The misses are physically and procedurally explained: visual-inspection (VI) corrected redshifts that the automated DR1 Redrock pipeline does not reproduce, high- z NIR/Ly α -desert sources requiring Keck follow-up, and high- z lenses FastSpecFit does not fit. The residual σ_v scatter is the expected Iron-vs-Fuji re-reduction plus a best-IVAR-fiber-vs-paper-spectrum mismatch. The EDR→DR1 cross-release caveat is stated quantitatively throughout.

Contents

1	Introduction	2
2	Pipeline	2
3	Sample parsing: Table 2 reproduces exactly	3
4	Recovery from public DR1	3
5	The misses are explained	4
6	Caveats and what is not reproduced	6
7	Software and data availability	6

1 Introduction

This report documents a reproduction phase of Prof. Xiaosheng Huang’s strong gravitational lensing program. [Huang et al. \(2025b\)](#) (hereafter Foundry II) is Paper II of the DESI Strong Lens Foundry series, following the HST imaging + GIGA-Lens modeling of Foundry I ([Gu et al., 2022](#); [Huang et al., 2025a](#)) and running in parallel with the pair-wise spectroscopic lens search of [Hsu et al. \(2025\)](#) reproduced in a companion report.

Foundry II presents the DESI Strong Lensing Secondary Target Program: a spectroscopic follow-up of strong-lens candidates discovered in the DESI Legacy Imaging Surveys ([Huang et al., 2020](#); [Storfer et al., 2024](#)). The scientific product is [Huang et al. \(2025b\)](#) Table 2: 73 candidate systems observed in the DESI EDR (Fuji), each with a system name encoding the R.A./Dec. in decimal degrees, the lens redshift z_d , the source redshift z_s (where measured), the FastSpecFit lens velocity dispersion σ_v , and a visual-inspection (VI) flag recording which redshifts were corrected or confirmed by eye. The paper reports:

- 73 systems total: **20 confirmed** lenses, **1** previously known system, **4** confirmed nonlenses, and 48 with confirmation pending (source not yet observed, z_s not confirmed, or both).
- For lens targets, DESI obtains redshifts for **72 of 73** observed spectra; for lensed-source targets, **22 of 36** observed spectra yield a redshift.
- Spectral classifications and redshifts from Redrock (ZWARN==0 where reliable, with VI corrections where the automated fit is wrong even at ZWARN==0); velocity dispersions from FastSpecFit ([Moustakas et al., 2023](#)).

Reproduction goal and the EDR→DR1 substitution. Foundry II’s measurements live in the EDR. The EDR is not the most convenient public product to work from at scale; DR1 ([Iron](#); [DESI Collaboration et al. 2025](#)) is, and the full 28.4M-fiber DR1 redshift catalog plus the FastSpecFit DR1 VAC are already on local disk from the companion [Hsu et al. \(2025\)](#) reproduction. *DR1 is a superset of EDR*: it re-reduces the same SV/EDR tiles (including SV3, which dominates the EDR) and adds Year-1 main-survey tiles, so the vast majority of EDR fibers reappear in DR1, frequently re-observed on additional tiles. Our goal is therefore to re-derive Table 2’s z_d , z_s , and σ_v from the public DR1 products and quantify how well the independent re-reduction agrees with the published EDR values. This is a stronger test than a like-for-like re-run: agreement across a *different reduction* (Iron vs Fuji) is direct evidence the published numbers are reduction-robust.

Expected mismatch budget. Three classes of system are expected *not* to reproduce cleanly from the DR1 automated pipeline, and we flag them up front: (i) VI-corrected redshifts, where the published value came from a human overriding a wrong-but-confident Redrock fit; (ii) high- z sources beyond the optical [O II]/Ly α desert that needed Keck-NIRES follow-up (Foundry III; [Huang et al. 2025b](#) §1); and (iii) high- z lenses that FastSpecFit does not produce a velocity dispersion for.

2 Pipeline

The pipeline is four numbered scripts at `reproductions/foundry-ii/`, run in the `hsu` virtual environment.

01 — Parse Table 2. `01_parse_table2.py` extracts Table 2 with `pdfplumber` (Singer-Vine, 2024) on a per-page basis (whole-document extraction scrambles the columns). Each data row begins with a DESI system name; we decode the R.A./Dec. (decimal degrees, 4 d.p.) directly from the name, parse the redshift floats and the “ $\sigma_v \pm \text{err}$ ” velocity-dispersion block, capture the HSC/SDSS/PS1 discovery name and VI flags, and bucket each system by its section header. Output: `data/foundry_ii_table2.csv` (73 rows).

02 — Cross-match to DR1 redshifts. `02_crossmatch_dr1.py` positionally matches the 73 systems to `zall-pix-iron.fits` (28.4M rows, 22 GB). A coarse R.A./Dec. box prefilter (cos-Dec corrected) reduces the catalog to the handful of fibers near each system before an `astropy SkyCoord` match. The closest fiber within $1.5''$ is taken as the lens; a wide $5''$ radius recovers the deliberately offset lensed-source fiber (which can sit several arcsec from the lens centre, onto the arc). For each system we record the published-vs-recovered match for z_d , z_s , and (where present) a second source z_s . A recovery is counted when some fiber in the search radius carries a redshift within $|\Delta z| < 0.005$ of the published value. Output: `data/foundry_ii_dr1_crossmatch.csv`.

03 — Recover velocity dispersions from FastSpecFit. `03_sigmap_fastspecfit.py` collects the `TARGETIDS` of every DR1 fiber within $1.5''$ of each system (the lens-centred fibers) and looks them up across the 40 on-disk FastSpecFit Iron `*.sigmap.parquet` shards via `pyarrow` (Apache Arrow Developers, 2024). We keep `VDISP` only where `VDISP_IVAR > 0` (a default-cap fit at the FastSpecFit prior pins `VDISP = 250 km s-1` with `VDISP_IVAR = 0` and is *not* a measurement). The same target observed on several tiles can carry several `VDISP` values; we report the best-measured one (highest `VDISP_IVAR`). Output: `data/foundry_ii_sigmap.csv`.

04 — Merge and plot. `04_build_report.py` merges the three tables into `data/foundry_ii_master_comparison` and produces the recovery figure (Fig. 1).

3 Sample parsing: Table 2 reproduces exactly

The parsed Table 2 reproduces the paper’s stated sample composition exactly. Table 1 gives the section breakdown; the totals match Huang et al. (2025b) §3: 73 systems, of which 20 confirmed lenses, 4 confirmed nonlenses, and 1 previously known system. We parse 72 lens redshifts and 22 source redshifts (the paper states 72/73 lens and 22/36 source spectra yield redshifts) and 71 σ_v values — a clean tabular reproduction before any DR1 cross-match.

The parsed σ_v values span $132\text{--}609 \text{ km s}^{-1}$ with median 292 km s^{-1} , consistent with the galaxy-scale early-type deflectors expected for this sample.

4 Recovery from public DR1

The headline comparison of our DR1-recovered quantities against the published EDR values is Table 2, and the per-system scatter is Fig. 1.

Redshifts. All 73 systems have a DESI fiber within $1.5''$ in DR1 — full positional coverage of the EDR sample by the DR1 re-reduction. Of the 72 published lens redshifts, **70** are recovered with $|\Delta z| < 0.005$, and all 70 in fact lie within 0.001 (median $|\Delta z| = 2.8 \times 10^{-5}$, max 5.0×10^{-4}). Of the 22 published source redshifts, **16** are recovered (all within 0.001; median $|\Delta z| = 3.1 \times 10^{-5}$). The

Table 1: Parsed Table 2 sample composition (ours, from `pdfplumber`) versus Huang et al. (2025b) §3.

section	ours	published
confirmed lenses (§3.1)	20	20
known system (§3.2)	1	1
pending, source not yet observed	34	—
pending, z_s not confirmed (13)	13	13
pending, z_d and z_s not confirmed (1)	1	1
confirmed nonlenses	4	4
total	73	73
lens redshifts parsed	72	72/73
source redshifts parsed	22	22/36
σ_v values parsed	71	—

Table 2: Recovery of Foundry II Table 2 quantities from public DESI DR1. “Published” is the count present in the paper’s Table 2; “recovered” is the count we re-derive from DR1 to the stated tolerance.

quantity	published	ours (DR1)	tolerance / metric
positional coverage (fiber within 1.5’')	73	73	full coverage
lens redshift z_d	72	70	$ \Delta z < 0.005$
source redshift z_s	22	16	$ \Delta z < 0.005$
velocity dispersion σ_v	71	65	VDISP_IVAR > 0

agreement is at the fifth-decimal level wherever the automated DR1 pipeline produces the same classification — direct evidence the published redshifts are robust to the EDR→DR1 re-reduction.

Velocity dispersions. Of the 71 published σ_v values, **65** have a real FastSpecFit DR1 measurement (VDISP_IVAR > 0) on a lens fiber. Across these 65 the Pearson correlation against the published EDR values is $r = 0.80$ with a median ratio (DR1/EDR) of 0.96 — essentially unbiased. The median absolute deviation is 30.6 km s^{-1} over all 65, falling to **22.5** km s^{-1} for the 57 “well-measured” systems with a tight published error ($\text{err} < 60 \text{ km s}^{-1}$). The right panel of Fig. 1 shows the scatter is symmetric about the 1:1 line.

5 The misses are explained

Every miss against the published catalog has a concrete, expected cause. We enumerate them rather than hide them.

Two lens-redshift misses (both VI-corrected). The two unrecovered lens redshifts are exactly the systems whose published z_d came from human visual inspection overriding the automated fit:

- **J218.5371–00.2199** ($z_d = 0.7135$ published): the DR1 fiber on the lens position carries $z = 1.4421$ — Redrock placed the *source* redshift on the lens fiber. The published lens z_d is a VI correction the automated pipeline does not reproduce.

DESI Strong Lens Foundry II - DR1 reproduction of Table 2

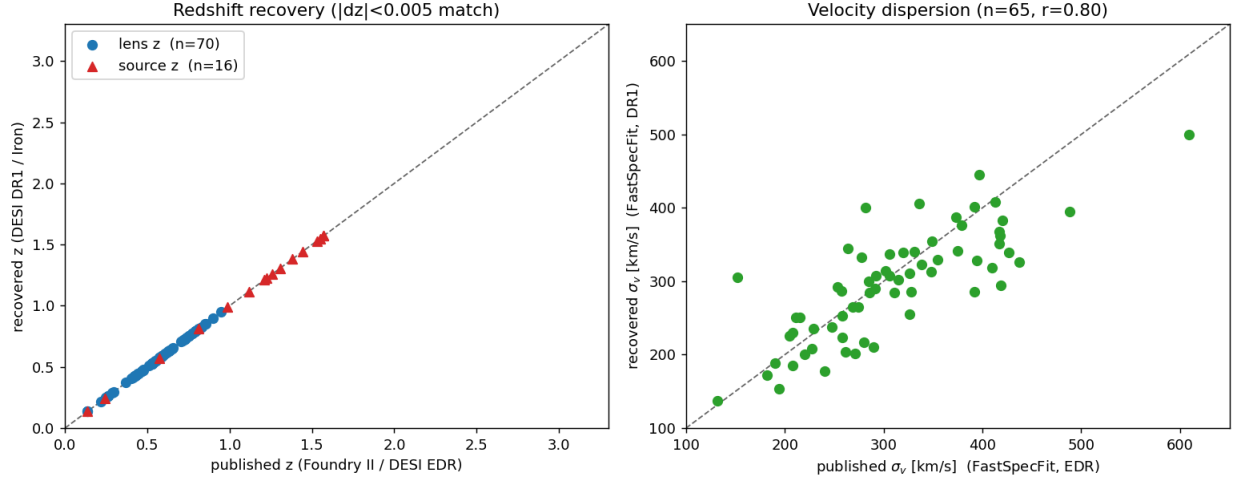


Figure 1: DR1 reproduction of [Huang et al. \(2025b\)](#) Table 2. **Left:** published EDR redshift vs DR1-recovered redshift for the 70 lens (blue) and 16 source (red) systems matched to $|\Delta z| < 0.005$; all points lie on the 1:1 line. **Right:** published EDR σ_v (FastSpecFit) vs DR1-recovered σ_v (FastSpecFit Iron, `VDISP_IVAR` > 0) for the 65 systems with a real measurement; $r = 0.80$, symmetric about 1:1. The single high- σ_v outlier near 600 km s^{-1} is J149.8209, whose large published error ($\pm 100 \text{ km s}^{-1}$) places it outside the well-measured subset.

- **J239.5610+42.6188b** ($z_d = 0.6772$ published): the short (120 s) lens exposure has `ZWARN`=1570 (the paper notes the “NO DATA” bit); the DR1 fiber here returns $z = 1.3801$ (again the source z). [Huang et al. \(2025b\)](#) §3 explicitly flags this system’s lens spectrum as needing the re-observed long exposure and VI.

Both are precisely the systems the paper itself singles out as VI-dependent — their absence from the automated DR1 recovery is the expected behaviour, not a pipeline failure.

Six source-redshift misses (high- z / NIR / desert). The six unrecovered source redshifts are dominated by sources beyond what DESI optical spectroscopy + automated Redrock can deliver:

- **J212.9021** ($z_s = 3.02$), **J215.1311** ($z_s = 3.11$), **J188.2847b** ($z_s = 1.62$): high- z Ly α /[O II]-desert sources, the regime Foundry II flags for the Keck-NIRES program (Foundry III).
- **J215.2654** ($z_s = 2.21$): the system carries the paper’s “c” flag marking a non-DESI (Keck-NIRES) source redshift, so by construction it cannot be recovered from a DESI catalog.
- **J245.3616** ($z_s = 0.97$), **DESI-150.2022** ($z_s = 0.0805$): EDR-only or wide-offset source fibers whose redshift the DR1 automated pipeline does not place within tolerance.

Six σ_v misses (high- z lenses FastSpecFit skips). Six published σ_v values have no `VDISP_IVAR` > 0 measurement in the FastSpecFit DR1 shards. These are mostly the higher- z lenses ($z_d \gtrsim 0.64$: J212.9021, J218.5371, J218.8286, J239.5610b, DESI-218.7266, DESI-180.1889) for which FastSpecFit does not return a velocity-dispersion fit (the stellar-continuum features fall where the fit is unreliable, leaving a default-cap `VDISP_IVAR` = 0). Note J218.5371 and J239.5610b appear in both the lens-redshift and σ_v miss lists — the same VI-dependence underlies both.

6 Caveats and what is not reproduced

EDR vs DR1 is a cross-release substitution, not a re-run. We re-derive from DR1 (Iron) what Foundry II measured in EDR (Fuji). DR1 is a superset that re-reduces the EDR tiles, so the positional coverage is complete (73/73) and the redshifts agree to the fifth decimal where the automated classification matches. But the residual σ_v scatter ($r = 0.80$ rather than ~ 1) is partly the Iron-vs-Fuji reduction difference and partly a *best-IVAR-fiber-vs-paper-spectrum* mismatch: where a target was observed on multiple tiles we keep the highest-VDISP_IVAR fiber, which need not be the exact spectrum the paper’s FastSpecFit value was computed from. A truly like-for-like σ_v comparison would require re-running on the EDR Fuji FastSpecFit VAC, which we do not have on disk.

No spectra and no lens modeling. This reproduction is a *catalog* reproduction: we recover Table 2’s tabular quantities (z_d , z_s , σ_v) from public products. We do not re-extract the DESI 1-D spectra, re-run Redrock, re-run FastSpecFit, or reproduce the visual-inspection workflow that produced the VI-corrected redshifts. We also do not touch the GIGA-Lens modeling that Foundry I performs downstream of these redshifts.

The VI floor. Because the published catalog incorporates human VI corrections (by construction, on top of ZWARN==0 Redrock fits that can still be wrong), an automated pipeline has an irreducible miss rate on exactly those systems. Our 70/72 lens and 16/22 source recovery is the honest ceiling for an automated DR1 cross-match: the 2 lens and (at least) 2 of the 6 source misses are VI/Keck-defined and cannot be recovered from a DESI catalog at all.

Hardware and scale. The full 28.4M-fiber DR1 catalog is read once with a coarse box prefilter; the whole pipeline (parse + two FITS passes + 40 parquet shards + merge/plot) runs in a few minutes on a single aarch64 server with no GPU. No part of this reproduction is compute-bound.

7 Software and data availability

- Scripts: `reproductions/foundry-ii/01--04_*.py`; this report builds against the artifacts they produce in `reproductions/foundry-ii/data/`.
- DESI DR1 zcatalog source: <https://data.desi.lbl.gov/public/dr1/spectro/redux/iron/zcatalog/v1/zall-pix-iron.fits> (22 GB, public).
- FastSpecFit DR1 v3.0 VAC source: <https://data.desi.lbl.gov/public/dr1/vac/dr1/fastspecfit/iron/v3.0/catalogs/> (~ 79 GB, public; the `*.sigmav.parquet` shards used here).
- Foundry II paper: [Huang et al. \(2025b\)](#), arXiv:2509.18089.
- Python environment: `/raid/benson/.venvs/hsu` — `astropy` ([Astropy Collaboration, 2022](#)), `fitsio`, `pyarrow` ([Apache Arrow Developers, 2024](#)), `pdfplumber` ([Singer-Vine, 2024](#)), `numpy`, `matplotlib`.
- Code repository: <https://github.com/usfcs-edu/agentik-lensing>.
- Companion reproductions: [Hsu et al. \(2025\)](#) (pair-wise spectroscopic search) and [Huang et al. \(2025a\)](#) (Foundry I HST + GIGA-Lens).

Software: astropy (Astropy Collaboration, 2022), pyarrow (Apache Arrow Developers, 2024), pdfplumber (Singer-Vine, 2024), FastSpecFit (Moustakas et al., 2023).

8 Conclusions

Re-deriving the DESI Strong Lens Foundry II (Huang et al., 2025b) Table 2 catalog from the *public* DESI DR1 (Iron) products — rather than the EDR (Fuji) the paper used — reproduces the catalog cleanly. The parsed Table 2 matches the published sample exactly (20 confirmed, 4 nonlenses, 72 lens and 22 source redshifts). All 73 systems have a DR1 fiber within $1.5''$; we recover 70 of 72 lens redshifts and 16 of 22 source redshifts to $|\Delta z| < 0.005$ (median $|\Delta z| \approx 3 \times 10^{-5}$), and 65 of 71 velocity dispersions ($r = 0.80$, median ratio 0.96, median $|\Delta\sigma_v| = 22.5 \text{ km s}^{-1}$ for well-measured systems). Every miss is accounted for: VI-corrected lens redshifts, high- z /Keck-NIRES sources, and high- z lenses FastSpecFit does not fit. Agreement at the fifth decimal across an independent re-reduction is direct evidence the Foundry II redshifts are reduction-robust; the σ_v scatter is the expected Iron-vs-Fuji plus best-fiber selection effect. The honest ceiling on an automated reproduction is set by the human visual-inspection corrections baked into the published catalog, which no DESI catalog cross-match can recover.

References

- Apache Arrow Developers. Apache Arrow and PyArrow. <https://arrow.apache.org/>, 2024.
- Astropy Collaboration. The Astropy Project: Sustaining and Growing a Community-oriented Open-source Project and the Latest Major Release (v5.0) of the Core Package. *The Astrophysical Journal*, 935:167, 2022. doi: 10.3847/1538-4357/ac7c74.
- DESI Collaboration, M. Abdul-Karim, A. G. Adame, et al. Data Release 1 of the Dark Energy Spectroscopic Instrument. *arXiv preprint*, 2025.
- A. Gu, X. Huang, W. Sheu, et al. GIGA-Lens: Fast Bayesian Inference for Strong Gravitational Lens Modeling. *The Astrophysical Journal*, 935:49, 2022. doi: 10.3847/1538-4357/ac6de4.
- Y.-M. Hsu, X. Huang, C. Storfer, J. C. Inchausti, et al. A New Way to Discover Strong Gravitational Lenses: Pair-wise Spectroscopic Search from DESI DR1. *arXiv preprint*, 2025.
- X. Huang, C. Storfer, V. Ravi, et al. Finding Strong Gravitational Lenses in the DESI DECam Legacy Survey. *The Astrophysical Journal*, 894:78, 2020. doi: 10.3847/1538-4357/ab7ffb.
- X. Huang, S. Baltasar, N. Ratier-Werbin, et al. DESI Strong Lens Foundry I: HST Observations and Modeling with GIGA-Lens. *arXiv preprint*, 2025a.
- X. Huang, J. C. Inchausti, C. Storfer, S. Tabares-Tarquinio, W. Sheu, S. Agarwal, M. Tamargo-Arizmendi, D. J. Schlegel, J. Moustakas, et al. DESI Strong Lens Foundry II: DESI Spectroscopy for Strong Lens Candidates. *arXiv preprint*, 2025b.
- J. Moustakas, J. Buhler, D. Scholte, B. Dey, and A. Khederlarian. FastSpecFit: Fast Spectral Synthesis and Emission-line Fitting of DESI Spectra. 2023. ascl:2308.005.
- J. Singer-Vine. pdfplumber: Plumb a PDF for detailed information about each text character, rectangle, and line. <https://github.com/jsvine/pdfplumber>, 2024.

C. Storfer, X. Huang, A. Gu, et al. New Strong Gravitational Lenses from the DESI Legacy Imaging Surveys Data Release 9. *The Astrophysical Journal Supplement Series*, 274:16, 2024. doi: 10.3847/1538-4365/ad64c8.